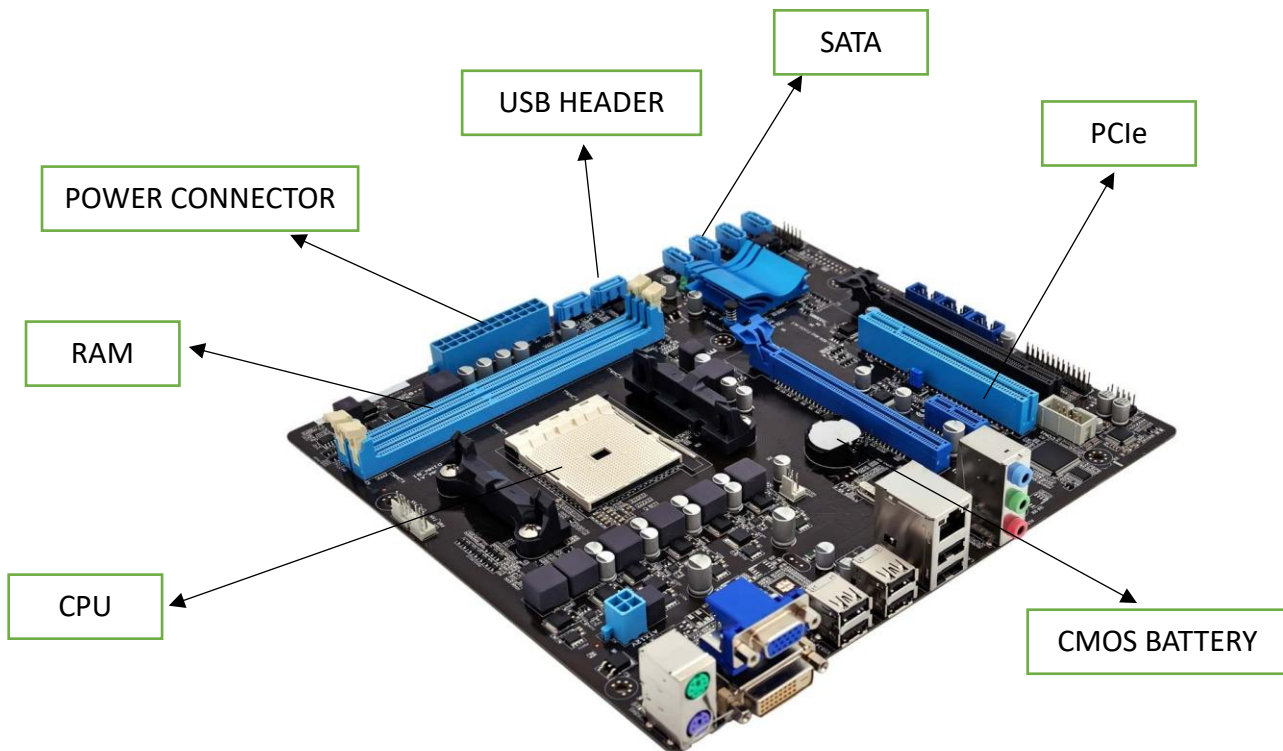


1. Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.)



2. Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

LGA1151 AND LGA1200 is no longer used in recent times.

Socket AM4

Pin grid array- processor contains tiny metal pins that fits perfectly into the holes on the motherboard. Released in 2017, it became one of the most long-lived and popular desktop platforms, supporting multiple generations of Ryzen processors (from 1000 to 5000 series) on the same motherboards. It utilizes DDR4 memory.

Compatible CPU- Ryzen 1000 through 5000 series

LGA 1700

This mainstream socket supports Intel's 12th, 13th, and 14th generation processors.

A compatible CPU model is the Intel Core i5-14400F.

LGA 1851

This is Intel's modern platform designed for current-generation processors. A compatible CPU model is the Intel Core Ultra 7 265K

3. Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

CPU- Often known as the brain of the computer, CPU executes instruction and process data for the entire system. It receives instructions and data from the computer's memory, interprets the instructions and determines the required task, executes it and finally sends the processed result back to the output device.

CHIPSET- Manages and control the flow of data between CPU, RAM Storage devices and peripherals. It works as traffic controller or nervous system ensuring the effective communication effectively.

- When we click on the app chipset detects the click then fetch the Spotify application data from the system.
- Chipset grabs data from the storage and passes it to the RAM then the CPU reads information and loads Spotify App user interface on the screen.
- Compressed audio packets over the internet is decoded into uncompressed digital audio by the CPU.
- Chipset manages this high speed data flow quickly translates it, safely sending it to CPU and RAM without any crashes.
- When CPU finishes data processing chipset sends data to digital to analog convertor which is built directly on the motherboard.
- As the CPU is fast, it handles billions of calculations in a second hence handling the app becomes easy and seamless when we are doing multiple things at once.

4. Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labelling each one.

